

Design and Production Engineering

EDPT 501 – “Engineering Design I”



Project

Prof. Dr. Hesham A. Hegazi
Professor of Engineering Design and CAD
Email: Hesham.Hegazi@guc.edu.eg
Office: C7.07

Shop Crane

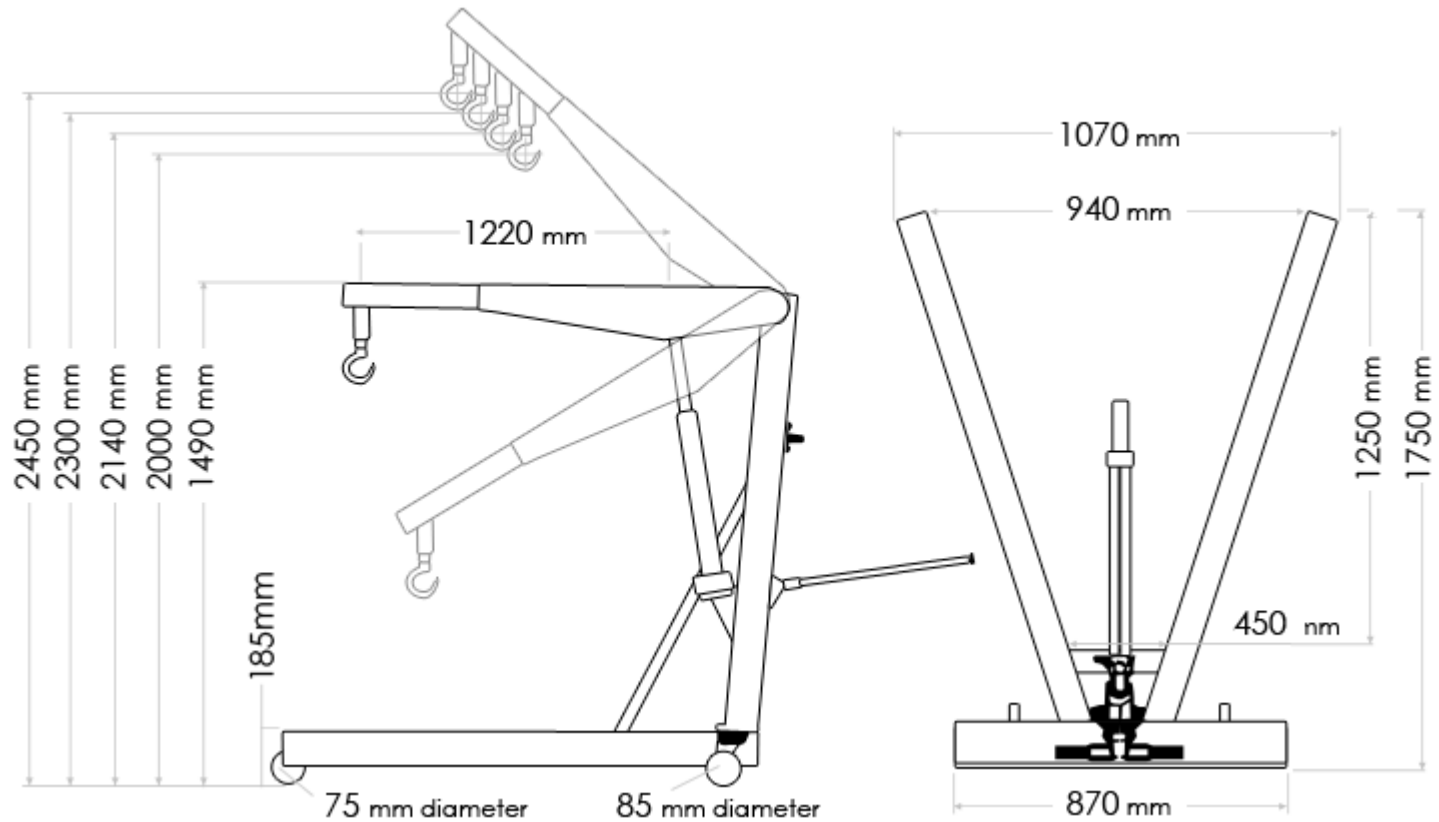


Different Designs



General Specifications

- Capacity: from 1 to 2 tons
- Dimensions



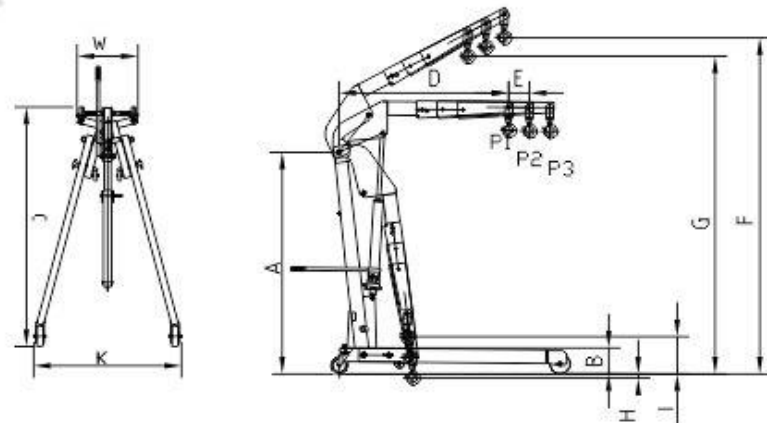
Example of Catalogues Information

ENGINE CRANE SC-B series



Model		SC500B	SC1000B	SC2000B
Capacity P1/P2/P3 (Kg)		500/425/350	1000/800/700	2000/1700/1500
Dimensions(mm)	A	1354	1597	1626
	B	165	90	208
	C	1582	1749	1911
	D	897	1231	1293
	E	102	150	150
	F	2080	2450	2490
	G	1920	2320	2330
	H	130	/	/
	I	330	280	250
	W	170	460	590
	K	960	1100	1170
Net Weight (Kg)		75	115	165

- Larger capacity than normal shop cranes at 3 position.
- Chrome plated lifting piston-rod and pump piston-rod.
- Relief valve to prevent overload.
- Forged heavy duty swivel hook.
- 25% overload testing before delivery.



Bolted Joints in Tension

SHOP CRANE



T1001Z



T1002G



T1002Z



T1003G

MODEL	Capacity (Ton)	Lifting Range (mm)	N.W. (kg)	Meas. (cm)	20'FCL (pcs)
T1001Z	1	110-2000	15 53	71x35x16 140x30x20	220
T1002G	2	280-2100	52 40	137x71x9 133x21x16	200
T1002Z	2	85-2000	70 22	145x30x18 80x58x9	220
T1003G	3	0-2200	29 72 18	133x86x11 142x30x21 73x20x18	110

Standard Components

Hydraulic Cylinder



Click image to open expanded view



5 Ton Ram for Engine Crane Hydraulic Ram Hoist Cylinder Jack Engine Lift

Brand: TECHTONGDA

We don't know when or if this item will be back in stock.

- Minimum height 642mm
- Maximum height 1110mm
- Adjustable height 500mm
- Maximum lifting weight 5T
- Material : Strong robust steel construction

› [See more product details](#)



Pro-Lift B-004D Grey
Hydraulic Bottle Jack - 4
Ton Capacity

Used in Cars

Standard Components

Wheels

[Back to results](#)



Roll over image to zoom in

4" Plate Caster Wheels Set of 4 1800LB with Screw Safety Dual Locking and Polyurethane Foam No Noise Wheels, Heavy Duty - 450 Lbs Per Caster

Brand: HOMIGRAND

★★★★★ 518 ratings | 16 answered questions

- SCREWS INCLUDE AND HEAVY DUTY: 360 degree rotation to facilitate rotation at any angle, Heavy duty: can withstand a single 450lb, a total of 1800lb weight
- SAFETY: The 4 wheels have brakes, ensuring the safety of people and goods at any time and at any angle
- CAPACITY: 550lbs each. Plate Size: 3.74" (95mm) x 2.6" (66mm). Hole center to hole center: 2.93" (74.5mm) x 1.8" (45mm). Mounting hole dimensions: 0.33" (8.5mm) x 0.46" (11.9mm)
- Wheel Dia.: 4" (100mm). Wheel Width: 1.25" (32mm). Wheel Material: Polyurethane. Overall Height: 5.15" (131mm)
- WARRANTY: 2 years warranty, quality problems, only change, no maintenance

Standard Components

Wheels



6" x 2" Heavy Duty Toolbox Caster Set of 4 with Phenolic Wheels, 4,800 lbs Capacity per Set of 4, 2 Swivels with Locking Brakes and 2 Rigid Casters, CasterHQ Brand

[Visit the CasterHQ Store](#)

★★★★★ 91 ratings

Amazon's **Choice** in Plate Casters by CasterHQ

Price: **\$129.99** + \$118.26 shipping

\$310.46 Shipping & Import Fees Deposit to Egypt [Details](#)

Available at a lower price from [other sellers](#) that may not offer free Prime shipping.

- WHEEL DIAMETER: 6" | WHEEL TREAD WIDTH: 2"
- LOAD CAPACITY: 1,200 LBS /EA / 4,800 LBS PER SET OF 4
- TOP PLATE SIZE: 4" X 4-1/2" | BOLT HOLE SPACING: 2-5/8" X 3-5/8" (Slotted to 3" x 3")
- OVERALL HEIGHT: 7-1/2" | BEARING TYPE: ROLLER BEARING
- MOUNTING BOLT SIZE: 3/8" (Mounting Hardware Not Included)

Price (Sample)

WOW ✂ **SUPER COUPON**

2 TON CAPACITY FOLDABLE SHOP CRANE



PITTSBURGH

Customer Rating
★★★★★

- Boom extends from 41" to 61-3/4"
- Crane height adjusts from 73-5/8" to 89"
- Clevis grab hook with safety latch
- Legs fold up for easy storage

**COMPARE TO
TEQ
\$239⁹⁹**
MODEL: 1404PV

**SAVE
\$60**
ITEM 60388
69514 shown

~~\$229⁹⁹~~ **\$179⁹⁹**

PRINT COUPON

BUY NOW


37575977
LIMIT 3 • VALID thru 1/20/19*

Standard of hollow steel sections

MILD STEEL HOLLOW SECTIONS

Metal Supplies™ have been supplying the UK building and construction industry with all types of structural steel for many years, including a wide range of Mild Steel Steel Hollow Sections: square, rectangular and circular.

Our mild steel tube, square and rectangular box section products come in a variety of grades, sizes and finishes and we also offer a quick turnaround cutting service, so if we don't stock the exact hollow steel section you're after, we can cut to order and ship your steel out to you in next to no time.



Our friendly team will be happy to discuss all of your hollow section requirements, so why not give us a call today on **0800 123 70 10** for advice and to get our latest prices. alternatively, drop us a line at: sales@metalsupplies.com

NOTE: To view all of our different Mild steel products (from beams to welded mesh) please navigate via the Products section at the top of the page.

Sizes/grades in stock for square, circular and rectangular hollow section:

MILD STEEL (SHS) SQUARE STEEL HOLLOW SECTION:

Generally available in 7.500 to 7.650 Metre lengths in Grade 43; however other lengths and grades are available. For your information Kg/m refers to the Kilograms per Metre of the section, as an example a 4.200 Metre length of 40 x 40 x 3mm (SHS) Square Steel Hollow Section (3.45 Kg/m) would weigh (4.2 x 3.45) 14.49 Kg's Total.

20 X 20 X 2MM (SHS) SQUARE STEEL HOLLOW SECTION (1.12 KG/M)
20 X 20 X 2.5MM (SHS) SQUARE STEEL HOLLOW SECTION (1.35 KG/M)
25 X 25 X 2MM (SHS) SQUARE STEEL HOLLOW SECTION (1.43 KG/M)
25 X 25 X 2.5MM (SHS) SQUARE STEEL HOLLOW SECTION (1.74 KG/M)
25 X 25 X 3MM (SHS) SQUARE STEEL HOLLOW SECTION (2.04 KG/M)
30 X 30 X 2MM (SHS) SQUARE STEEL HOLLOW SECTION (1.68 KG/M)
30 X 30 X 2.5MM (SHS) SQUARE STEEL HOLLOW SECTION (2.14 KG/M)
30 X 30 X 3MM (SHS) SQUARE STEEL HOLLOW SECTION (2.51 KG/M)
40 X 40 X 2MM (SHS) SQUARE STEEL HOLLOW SECTION (2.31 KG/M)
40 X 40 X 2.5MM (SHS) SQUARE STEEL HOLLOW SECTION (2.92 KG/M)
40 X 40 X 3MM (SHS) SQUARE STEEL HOLLOW SECTION (3.45 KG/M)

MILD STEEL (RHS) RECTANGULAR STEEL HOLLOW SECTION:

Generally available in 7.500 to 7.650 Metre lengths in Grade 43; however other lengths and grades are available. For your information Kg/m refers to the Kilograms per Metre of the section, as an example a 4.200 Metre length of 50 x 30 x 3mm (RHS) Rectangular Steel Hollow Section (3.45 Kg/m) would weigh (4.2 x 3.45) 14.49 Kg's Total.

40 X 25 X 2MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (1.83 KG/M)
40 X 25 X 2.5MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (2.23 KG/M)
50 X 20 X 2MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (1.96 KG/M)
50 X 25 X 2.5MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (2.62 KG/M)
50 X 25 X 3MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (3.07 KG/M)
50 X 30 X 2.5MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (2.92 KG/M)
50 X 30 X 3MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (3.45 KG/M)
50 X 40 X 3MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (3.77 KG/M)
60 X 30 X 3MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (3.77 KG/M)
60 X 40 X 2.5MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (3.71 KG/M)
60 X 40 X 3.2MM (RHS) RECTANGULAR STEEL HOLLOW SECTION (4.66 KG/M)

Input Data:

Capacity: 1 Ton

Lifting Range: 110 to 2000 mm

Height (if horizontal) = 1500 mm Horizontal Length
(Approx.) = 1700 mm

Wheel Diameter (Approx.) = 100 mm

Sequence of Design

- Literature review (review of different types of cranes)
- Defining Capacity and overall dimensions
- Motion Study
(to find the dimensions of members)
- Force Analysis
(to specify the critical section for each member)
- Stresses calculations
(for each member – SF = 1.3 minimum)
- Finding Parts dimensions
(Use standard cross-sections)
- Draw a 3D model of the Crane using SolidWorks
- Draw 2D assembly drawing of the Crane.



Thank You